

Wenhui Li

Ph.D.

📍 Indiana University ✉ wenhuili324@gmail.com ☎ (+1) 812-671-8803 in LinkedIn G Google Scholar

Summary

- 10+ years of expertise in multiscale modeling and simulation across atomistic, mesoscopic, and continuum scales
- Skilled in physics-informed machine learning models across a range of scientific and engineering domains
- Proven record of scientific innovation with 27 publications (800+ citations) and interdisciplinary collaborations

Skills

- **Domain Knowledge:** Computational Soft Materials, Solid/Fluid Mechanics, Interface Science
- **Programming:** Python (PyTorch, TensorFlow, pandas, scikit-learn, SciPy), C, C++, Bash
- **Machine learning:** LLM, Gen AI (GAN & VAE), GNN, Unsupervised learning
- **Simulation & Modeling:** Atomistic (Molecular Dynamics (MD) & Monte Carlo (MC)), Mesoscopic (Coarse-Grained Modeling), Continuum (Finite Difference/Volume/Element Methods)
- **Tools & Platforms:** HPC (SLURM), Linux, Git, AWS
- **Soft Skills:** Fast Learner, Detail-oriented, Multitasker, Collaborator, Professional Communicator

Professional Experience

Postdoctoral Researcher

Sep. 2022 – Present

Indiana University, Bloomington | Toyota, Princeton, IN, USA

- Developed Graph Neural Networks to accelerate MD simulations, improving computational speed by ~30%
- Built and deployed ML-based surrogate models for Toyota's paint defect prediction, achieving a ~50% reduction in manual inspection
- Contributed to the AI-based energy analytics project (AnalytiXIN) for Indiana State manufacturers, reducing energy consumption by 20–30%
- Partnered with experimentalists to elucidate nanoparticle encapsulation mechanisms via coarse-grained MD and PMF analysis, contributing to the rational design of cargo loading encapsulation
- Uncovered molecular shear thinning mechanisms in lubricants via MD simulations and ML-assisted analysis

Postdoctoral Researcher

Jul. 2021 – Jul. 2022

University of Alberta, Edmonton | Suncor Energy & Syncrude, Calgary, AB, Canada

- Simulated water/oil/asphaltene interfaces under electric fields, revealing field-induced disruption of asphaltene $\pi - \pi$ stacking and adsorption, with salt ions enhancing interfacial stability
- Investigated brine droplet behavior in oil/asphaltene under electric fields via MD simulations, showing that ion type, concentration, and asphaltene adsorption jointly govern droplet stability and migration—offering insights into electrostatic demulsification mechanisms

PhD Student

Aug. 2017 – Jul. 2021

University of Alberta, Edmonton, AB, Canada

- Investigated how surfactants, co-surfactants, salt ions, and nanoparticles affect the structural and dynamic properties of hydrocarbon–water interfaces, highlighting the synergistic role of co-surfactants in reducing interfacial tension and clarifying the debated effects of nanoparticles and ion concentrations
- Investigated CO₂ solubility in confined environments, revealing how surface chemistry, pore size, and salt ions jointly govern “over-solubility” or “under-solubility,” demonstrating the potential of CO₂ geo-sequestration
- Developed in-house codes to characterize amorphous nanoporous media and simulate gas transport, uncovering how porosity, pore connectivity, size, and tortuosity influence diffusion—providing insights for optimizing shale gas recovery.

Master Student

Sep. 2014 – Jul. 2017

China University of Geosciences at Beijing, Beijing, China

- Developed and validated a mathematical model for foamy heavy oil transport in porous media, demonstrating

superior performance compared to existing models

- Constructed and validated a mathematical model capturing the temporal evolution of pressure fields with source/sink terms

Education

Ph.D. in Civil and Environmental Engineering <i>University of Alberta</i> , Edmonton, AB, Canada	<i>Jul. 2021</i>
M.S. in School of Energy <i>China University of Geosciences</i> , Beijing, China Outstanding Graduate Thesis (top 5%)	<i>Jul. 2017</i>
B.S. in School of Energy <i>China University of Geosciences</i> , Beijing, China Graduated with Distinction (Ranked 3/67)	<i>Jul. 2014</i>

Selected Publications & Presentations

Authored 27 peer-reviewed publications, including 12 as first author, with 800+ citations

Delivered 15 presentations at leading conferences, including the APS, AIChE, and Society of Rheology
(Partial list shown; full list available on my [Google Scholar](#) )

- [Industrial Energy Forecasting Using Dynamic Attention Neural Networks](#).  *Energy and AI*, 20 (2025): 100504.
- [Data-Driven Discovery of Molecular Structure Patterns in Shear Thinning of Lubricants](#).  *95th Society of Rheology*, Austin, TX, USA, Oct. 14-17, 2024.
- [Electrostatics-Driven Peptide-Directed Encapsulation of Nanoparticles into Protein Cages](#).  *APS March Meeting*, Minneapolis, MN, USA, Mar. 3-8, 2024.
- [Dependence of Methane Transport on Pore Informatics in the Amorphous Nanoporous Kerogen Matrix](#).  *Langmuir*, 40 (2023): 687-695.
- [Molecular Dynamics Simulation on Water/Oil/Asphaltene Interface Subjected to Electric Field](#).  *Journal of Colloid and Interface Science*, 628 (2022): 924-934.

Leadership & Service

- Led the project “Interfacial Behaviors During the Development of Unconventional Gas Reservoirs”; authored the research proposal and secured USD 4,000 in funding from May 2019 to May 2020.
- Mentoring 5 graduate/undergraduate students, resulting in joint publications and successful project delivery
- Co-chaired the “Data-driven Rheology” session at *Society of Rheology Meeting*, 2024
- Reviewed 20+ papers for *Physical Review Letters*, *Langmuir*, *Fuel*, *Chemical Engineering Journal*, etc.
- Guest Editor for *Geofluids* special issue on Multiscale and Multiphysics Approaches to Fluids Flow in Unconventional Reservoirs, 2022

Honors & Awards

Future Energy Systems Travel Fund, University of Alberta	<i>2019</i>
Graduate Student Association Academic Travel Grant, University of Alberta	<i>2018</i>
Graduate Recruitment Scholarship (CAD 10,000), University of Alberta	<i>2017</i>
National Scholarship (2×, top 3%, CNY 20,000), Ministry of Education, China	<i>2015 & 2016</i>